

WEEK 4 PROGRESS REPORT

WHO DEFINES THE PROBLEM? A COMPUTATIONAL ANALYSIS OF HOW EU INSTITUTIONS FRAME ARTIFICIAL INTELLIGENCE REGULATION

20 September 2026

Project GitHub Link: <https://github.com/michelle-qyh/SOSC314-Course-Project/>

Division of Responsibilities

For this project stage, each group member was responsible for the following components:

Tim Maurice Steiner:

Primary responsibility: Operationalisation and analytic design — frame dictionary, representation specifications, comparison of approaches, written deliverables.

Specific tasks completed this week: Built the representation and measurement modules (represent.py, measures.py) and the analysis driver (03_operationalisation.py); recovered Parliament adopted texts via the Open Data Portal (ep_portal.py); wrote the progress report and figure.

Yuhan Qiu:

Primary responsibility: Operationalisation via unsupervised topic modelling (LDA and NMF) as a cross-check against the dictionary and Fightin' Words routes

Specific tasks completed this week: Write “04_topic_modeling.ipynb”, preprocessing pipeline including two rounds of boilerplate detection and removal, coherence-based K selection, dual-representation topic modelling, paragraph- vs document-weighted robustness check, face validity check against representative paragraphs

Corpus Update

Week 3 flagged Parliament coverage as the main threat to comparability and undertook to attempt recovery. The Parliament website blocks automated access, but the Parliament Open Data Portal exposes adopted texts through a public API. Matching CELLAR works to Portal records on title fails — CELLAR stores the long formal title while the Portal stores a short label, giving similarity scores around 0.03 — and date alone is ambiguous, since Parliament adopts dozens of texts per sitting day. Date combined with the EuroVoc artificial-intelligence subject tag is decisive: five of seven adopted texts were uniquely identified, including Parliament's legislative resolution on the AI Act. The remaining two fall on days with no or two AI-tagged texts and yield no match rather than a guess. The corpus is now 139 documents, 27,732 paragraphs and 2.04 million tokens; Parliament coverage rises from 45 to 52 per cent.

Operationalisation

A representation is fixed by four decisions: how a string becomes tokens, whether unigrams or also bigrams, raw counts or tf-idf weighting, and a minimum document frequency. Tokenisation lowercases, keeps alphabetic tokens of at least three characters, and applies up to three stopword lists: ordinary English function words; EU procedural furniture (regulation, article, paragraph, Commission, Parliament, Council, rapporteur); and institutional rhetorical verbs (calls, stresses, underlines, whereas, having regard). The second and third lists are the substantive choices, and the report treats them as variables rather than defaults.

Three measures are built on that. Dictionary scoring operationalises the seven frames of the project proposal as term lists and counts matches per 100 tokens, aggregating paragraphs to documents and documents to institutions; profiles are reported as each institution's share of its own total frame signal, which removes the

AI-density differences already described in Week 3 and leaves relative emphasis. Fightin' Words (Monroe, Colaresi and Quinn 2008) computes log-odds ratios with an informative Dirichlet prior, so that rare terms are shrunk toward no difference and the ranking reflects both the size of a difference and the confidence in it. Topic modelling imposes no categories either, but clusters the same representation into latent themes rather than ranking individual terms — fit as LDA on raw counts and NMF on tf-idf, with K selected by coherence over {5,8,10,12} and vocabulary held fixed across both so the comparison isolates the count-versus-weighting decision; two paragraph-level formatting artifacts (figure captions, legislative preamble formulas) were identified and filtered before fitting. The three differ in where the categories come from: dictionary scoring imposes the hypotheses' own categories; Fightin' Words imposes none but returns single word, and topic modelling also imposes none but returns multi-word themes, at the cost of requiring a human label for each.

What the Representation Choice Does

The controlled comparison is the week's clearest result, and it cuts against the data-driven measure. Under the baseline specification, the terms most distinctive of Parliament against the Council are regard, calls, whereas, having, stresses and highlights; the Council's are article, regulation, annex, paragraph and section. These are not frames. They are the grammar of a resolution against the grammar of a legal text — the method recovers document genre and institutional house style, which it can do almost perfectly, and which tells us nothing about how AI is construed.

Removing procedural terms and then rhetorical verbs changes the picture substantively rather than cosmetically. Parliament's distinctive vocabulary becomes technologies, human, ethical, research, digital and development; the Council's becomes conformity, harmonisation, testing, legislation and office; the Commission's becomes cloud, liability, costs, computing and adoption. Only at this point does the output speak to the research question.

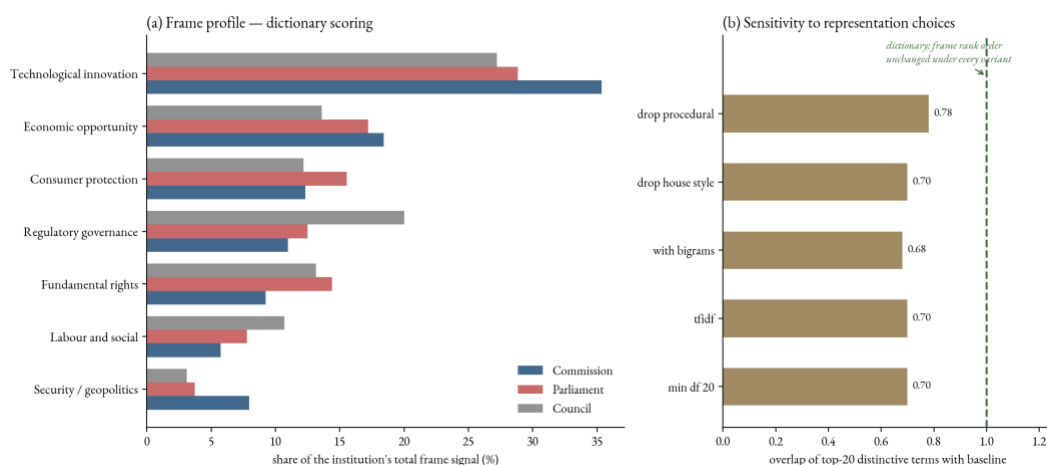


Figure 1. Panel (a): frame profile by institution under dictionary scoring, as each institution's share of its own total frame signal. Panel (b): sensitivity of Fightin' Words to representation choices, measured as overlap of the top-20 distinctive terms with the baseline specification; the dictionary's frame rank order is unchanged under all six.

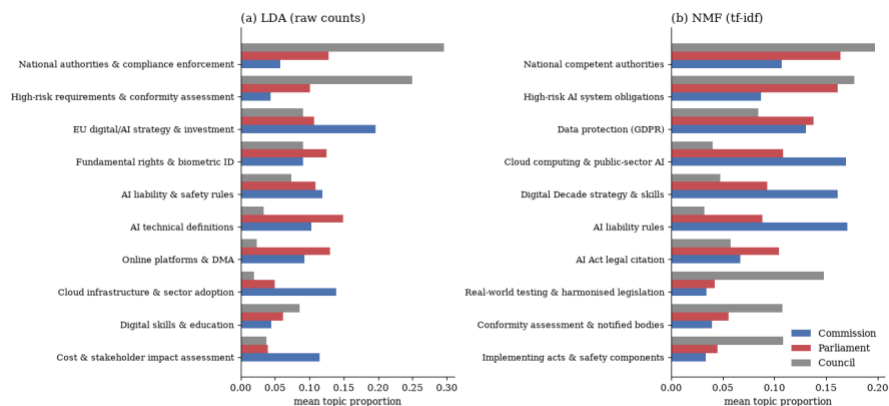


Figure 2. Topic proportion by institution under two representations: (a) LDA on raw counts, (b) NMF on tf-idf. Council's two largest topics — enforcement/compliance authorities and high-risk conformity assessment — jointly account for over half its paragraph share, consistent with its lead on regulatory governance in Figure 1a

Comparing the Two Approaches

Measured against the baseline, the top-20 distinctive terms retain only 0.68 to 0.80 overlap across the six specifications (Figure 1b): roughly a quarter to a third of what counts as most distinctive is a property of the preprocessing rather than of the institutions. The dictionary measure moves by 0.38 percentage points on average and 1.08 at most, and the rank order of frames within each institution is identical under every variant. The difference is structural, not incidental. Fightin' Words ranks an open vocabulary, so terms near the threshold cross it easily; the dictionary aggregates many terms into seven categories, and aggregation absorbs the variation. The trade-off is equally structural: the dictionary is stable because it is blind outside its own lists, and its apparent robustness is partly a restatement of the assumption that the seven frames are the right categories.

Where the two agree, the agreement is informative because the routes are independent. The Council's distinctive vocabulary — conformity, harmonisation, testing, legislation — corresponds to the frame on which it scores highest, regulatory governance at 20.0 per cent of its frame signal against 11.0 for the Commission. Parliament's human, ethical and protection correspond to its leads on fundamental rights (14.4 per cent) and consumer protection (15.6). The Commission's cloud, computing and costs correspond to its lead on technological innovation (35.4 per cent) and economic opportunity (18.4).

A third, fully unsupervised route — topic modelling via LDA and NMF, imposing no categories at all — converges on the same picture. Council's two most distinctive topics (enforcement/compliance authorities, 29.6%; high-risk conformity assessment, 24.9%) jointly account for over half its paragraph-level content, consistent with its lead on the dictionary's regulatory governance frame and its Fightin' Words vocabulary (conformity, harmonisation, testing). The three routes differ in where their categories come from — imposed by us, ranked from an open vocabulary, or inferred by a model — and in modelling family, but they are not independent: all run on the same cleaned corpus with broadly similar tokenisation, and the topic route adds a boilerplate pass the other two do not have. Given that representation choices demonstrably move the Fightin' Words ranking, convergence here is better read as three different category schemes recovering the same underlying emphasis than as independent confirmation. The Council estimate also rests on the least data — 4,819 paragraphs against the Commission's 17,695 — so its concentration is the least precisely measured of the three.

Substantive Interpretation

The frame profile supports H3 and qualifies H2 and H4. Parliament does lead on fundamental rights and consumer protection, as expected. The Commission leads on technological innovation and economic opportunity, but not on regulatory governance — where it scores lowest of the three, against the expectation in H2 that the initiating and implementing institution would emphasise standards and oversight. H4 fares worse: the Council was expected to emphasise security and competitiveness, and it is the lowest of the three on both (3.1 and 13.6 per cent), while leading decisively on regulatory governance. The reading we find most plausible is that these profiles track institutional function rather than institutional preference. The Commission writes the case for acting and the technical substance; the Council writes the machinery of implementation its member states will have to operate; Parliament writes the reasons a citizen should care. On that reading the security frame is thin everywhere because the AI Act is internal-market legislation, and H4 imported an expectation about the Council's general politics into a file where it does not apply.

The genre control shows how far this is provisional. Restricted to formal positions, the Council's governance share falls from 20.0 to 4.5 per cent and its labour and social share rises from 10.7 to 26.2 — but that subset is seven documents, too few to support either number. What the check establishes is not a corrected estimate but that the Council result is carried by its legislative notes, which is precisely where the genre confound was expected to bite.

Limitations and Next Steps

The dictionary is ours and unvalidated: no hand-coded sample has yet tested whether a paragraph scored high on a frame reads that way. That validation is the next step, and until it is done the frame profile is a measurement, not a finding. Term lists also overlap — jobs appears under both economic opportunity and labour, risk under both consumer protection and governance — so the frames are not independent. Both the dictionary and the topic model remain unvalidated against ground truth: no hand-coded sample has tested whether a paragraph scored high on a frame — or assigned high weight to a topic — actually reads that way at the level a human coder would agree with. Week 5 will hand-code a stratified sample of roughly 200 paragraphs and report agreement against **both** the dictionary and the topic assignments, providing a shared validation benchmark across all three operationalisations.